TechBridge Sol

Quality Controle Policy

Effective Date: January 2026
Applies To: All Engineering, QA, and Development Staff

1. Purpose

The purpose of this Quality Control / Quality Assurance Policy is to ensure that all software products developed at TechBridge Sol meet **high-quality, reliable, and secure standards**. This policy defines roles, responsibilities, processes, and metrics for quality control throughout the software development lifecycle (SDLC).

All engineers, QA staff, and project stakeholders are required to comply with this policy.

2. Scope

This policy applies to:

- Internal tools, SaaS applications, APIs, and microservices
- Web and mobile applications
- System integrations and third-party services
- Software development, deployment, and maintenance activities

The policy covers all aspects of quality, including functional, non-functional, security, and performance testing.

3. Core Quality Principles

All deliverables must adhere to these principles:

- **Correctness:** Software must function as intended and meet requirements
- **Reliability:** Systems must operate consistently under expected conditions
- **Maintainability:** Code must be modular, documented, and easy to modify
- **Performance:** Applications must meet performance benchmarks and scalability requirements
- **Security:** Software must protect sensitive data and prevent unauthorized access

4. Roles and Responsibilities

- **QA Lead / Manager:** Oversees QA strategy, ensures compliance, and approves releases
- **QA Engineers:** Plan, execute, and document tests; identify defects; ensure standards
- **Developers:** Follow coding standards, write unit tests, and fix defects promptly
- **Project Manager:** Coordinates QA schedules, ensures resources, and signs off on quality metrics

All team members share responsibility for maintaining quality.

5. Quality Assurance Processes

5.1 Code Reviews

- Mandatory for all code changes
- Review for adherence to coding standards, security, and functionality
- Feedback must be addressed before merging

5.2 Automated Testing

- Unit, integration, and regression tests required for each feature
- Continuous integration pipelines must pass before code merges
- Automated test coverage metrics should meet defined thresholds

5.3 Manual Testing

- Exploratory testing for edge cases and usability issues
- User Acceptance Testing (UAT) for stakeholder validation
- Document test results and defects clearly

6. Quality Control Measures

- Maintain **checklists and templates** for all QA activities
- Track metrics such as defect density, test coverage, and issue resolution time
- Conduct peer reviews of test plans and results
- Use version-controlled repositories for test scripts and documentation

7. Issue Management

- All defects must be logged in the approved issue tracking system
- Severity and priority assigned according to company standards
- QA engineers and developers collaborate to resolve defects promptly
- Defects must be verified after resolution before closure

8. Release Quality Gate

Before any release to staging or production:

- All critical and high-severity defects must be resolved
- Regression suite must pass in the target environment

- QA Lead or Project Manager must sign off on quality metrics
- Documentation of test results and release notes must be completed

No release may proceed without meeting the defined quality gate criteria.

9. Continuous Improvement

- Conduct **post-release reviews** to identify lessons learned
- Update test cases, QA processes, and documentation accordingly
- Encourage team feedback and adopt best practices to improve overall quality
- Track KPIs to measure improvements in defect rates, coverage, and cycle time

10. Policy Compliance

Non-compliance may result in:

- Deployment of defective or insecure software
- Increased production incidents or client issues
- Escalation to engineering management for corrective measures

Adherence ensures high-quality deliverables and customer satisfaction.

11. Policy Review and Updates

This QA Policy will be reviewed annually or whenever there are changes in technology, development practices, or business needs. Engineers and QA staff are expected to comply with the latest version.